

Nikhil Varma Keetha

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Education

Carnegie Mellon University | Ph.D. in Robotics

Aug 2022 - Present

Advisors: [Sebastian Scherer](#) and [Deva Ramanan](#)

Thesis: [Scaling Representation Learning for Spatial Intelligence via Active Implicit Memory](#)

Carnegie Mellon University | M.S. in Robotics

Aug 2022 - May 2024

Advisor: [Sebastian Scherer](#) | GPA: 4.12/4.0

Thesis: [Splat, Track & Map 3D Gaussians for Dense RGB-D SLAM](#)

Indian Institute of Technology (ISM) Dhanbad

Aug 2018 - May 2022

B.Tech. in Engineering Physics, with Distinction | GPA: 8.94/10.0

Experience

Meta Reality Labs

Zurich / Pittsburgh

Visiting Researcher

Aug 2024 - Present

Research Scientist Intern

May 2024 - Aug 2024

- Research on generalizable 3D reconstruction and scene generation with [Peter Kotschieder](#).
- Contributions include MapAnything, FlowR, Pomelo, and Meta Hyperscape.

AirLab, Carnegie Mellon University

Jun 2022 - Present

Graduate Research Assistant | Pittsburgh, PA

- Research on generalizable spatial perception with Sebastian Scherer and Deva Ramanan, including topics such as large-scale representation learning, visual correspondence, 3D/4D understanding, and spatial memory.
- Contributions include AnyLoc, SplatAM, UFM, Any4D, and perception systems for aerial and off-road navigation, semantic exploration, and collision avoidance.

Carnegie Mellon University

Apr 2021 - May 2022

Robotics Institute Summer Scholar (RISS) | Pittsburgh, PA

- Developed AirObject, a temporal graph embedding model for object identification (CVPR 2022), with Chen Wang and Sebastian Scherer.

Mila, Robotics and Embodied AI Lab

Jan 2021 - Jul 2021

Research Intern | Montreal, Canada

- Studied multi-view representation learning, differentiable mapping, and NeRF-based representations (RSS 2023) with Krishna Murthy Jatavallabhula and Liam Paull.

QUT Centre for Robotics

Aug 2020 - Feb 2021

Undergraduate Research Intern | Brisbane, Australia

- Developed a hierarchical model of environment- and place-specific utility for visual place recognition (IEEE RA-L 2021), with Sourav Garg and Michael Milford.

PURE EV

May 2020 - Aug 2020

R&D Data Analyst | Hyderabad, India

- Analyzed vehicle and battery sensor data and developed range-prediction frameworks with Nishanth Dongari.

IIT (ISM) Dhanbad

Nov 2019 - Aug 2020

Undergraduate Research Intern | Dhanbad, India

- Developed U-Det, a U-Net-based model for lung nodule segmentation, with A.C.S. Rao.

Awards**Meta Academic Research Grants**

2025 - 2027

Thesis research at CMU supported by two Meta Academic Research Grants (\$400K each).

Outstanding Reviewer, ECCV

2026

Outstanding Reviewer, CVPR

2026

Outstanding Reviewer, ICCV

2025

Outstanding Reviewer, CVPR

2025

Robotics Institute Summer Scholar

2021

Carnegie Mellon University

Diversity Grant, ACCV

2020

Featured ResearchSelected work, latest public release first. For the complete publication list, see [Google Scholar](#). * Equal contribution.**Pomelo: Multi-Perspective Generative Model**

Meta Reality Labs

Public research demo · 2026

[Thread](#)

A flexible generative model that can solve multiple tasks in one go: novel view synthesis, 3DGS reconstruction, camera-controlled flythroughs and 360s.

Any4D: Unified Feed-Forward Metric 4D ReconstructionJay Karhade, **Nikhil Keetha**, Yuchen Zhang, Tanisha Gupta, Akash Sharma, Sebastian Scherer, Deva Ramanan

CVPR 2026

[Project](#) / [Paper](#) / [Code](#) / [Demo](#)

A feed-forward model that jointly predicts metric 3D geometry and dense motion across multiple frames, with flexible support for additional sensor inputs.

RAVEN: Resilient Aerial Navigation via Open-Set Semantic Memory and Behavior AdaptationSeungchan Kim, Omar Alama, Dmytro Kurdydyk, John Keller, **Nikhil Keetha**, Wenshan Wang, Yonatan Bisk, Sebastian Scherer

ICRA 2026 · Oral

[Project](#) / [Paper](#) / [Code](#) / [Video](#)

Persistent 3D semantic memory and adaptive search behaviors enable aerial robots to find objects across large, unstructured outdoor environments.

MapAnything: Universal Feed-Forward Metric 3D Reconstruction

Nikhil Keetha, Norman Müller, Johannes Schönberger, Lorenzo Porzi, Yuchen Zhang, Tobias Fischer, Arno Knapitsch, Duncan Zauss, Ethan Weber, Nelson Antunes, Jonathon Luiten, Manuel Lopez-Antequera, Samuel Rota Bulò, Christian Richardt, Deva Ramanan, Sebastian Scherer, Peter Kotschieder

3DV 2026

[Project](#) / [Paper](#) / [Code](#) / [Model](#) / [Demo](#)

A unified transformer model that directly regresses the factored metric 3D geometry of a scene given various types of inputs (images, calibration, poses, or depth).

UFM: A Simple Path towards Unified Dense Correspondence with Flow

Yuchen Zhang, **Nikhil Keetha**, Chenwei Lyu, Bhuvan Jhamb, Yutian Chen, Yuheng Qiu, Jay Karhade, Shreyas Jha, Yaoyu Hu, Deva Ramanan, Sebastian Scherer, Wenshan Wang
NeurIPS 2025

[Project](#) / [Paper](#) / [Code](#) / [Model](#) / [Demo](#)

One transformer unifies optical flow and wide-baseline matching, learning general-purpose dense correspondence from their combined training data.

Demonstrating ViSafe: Vision-enabled Safety for High-speed Detect and Avoid

Parv Kapoor*, Ian Higgins*, **Nikhil Keetha***, Jay Patrikar*, Brady Moon, Zelin Ye, Yao He, Ivan Cisneros, Yaoyu Hu, Changliu Liu, Eunsuk Kang, Sebastian Scherer
RSS 2025 · Demonstrations

[Project](#) / [Paper](#) / [Video](#)

A vision-only detect-and-avoid system which integrates high-speed perception with safety-aware control. Demonstrated real flight tests with closure rates up to 144 km/h.

RayFronts: Open-Set Semantic Ray Frontiers for Online Scene Understanding and Exploration

Omar Alama, Avigyan Bhattacharya, Haoyang He, Seungchan Kim, Yuheng Qiu, Wenshan Wang, Cherie Ho, **Nikhil Keetha**, Sebastian Scherer
IROS 2025

[Project](#) / [Paper](#) / [Code](#)

An open-set semantic mapping system which combines 3D voxels with ray frontiers, letting robots reason about and explore beyond their depth-sensing range.

FlowR: Flowing from Sparse to Dense 3D Reconstructions

Tobias Fischer, Samuel Rota Bulò, Yung-Hsu Yang, **Nikhil Keetha**, Lorenzo Porzi, Norman Müller, Katja Schwarz, Jonathon Luiten, Marc Pollefeys, Peter Kotschieder
ICCV 2025 · Highlight

[Project](#) / [Paper](#)

Learning a direct mapping between incorrect renderings and their corresponding ground-truth images, augmenting scene captures with consistent novel, generated views to improve reconstruction quality.

Meta Hyperscape

Meta Reality Labs
Public demo · 2024 / Capture launch · 2025

[Project](#)

Photorealistic reconstruction turning captured physical spaces into immersive digital replicas that people can explore and share in VR.

Map It Anywhere (MIA): Empowering Bird's Eye View Mapping using Large-scale Public Data

Cherie Ho*, Jiaye Zou*, Omar Alama*, Sai Mitheran Jagadesh Kumar, Benjamin Chiang, Taneesh Gupta, Chen Wang, **Nikhil Keetha**, Katia Sycara, Sebastian Scherer
NeurIPS 2024

[Project](#) / [Paper](#) / [Code](#)

A data engine pairing street-level imagery with public maps at scale, enabling more generalizable bird's-eye-view mapping across diverse environments.

SplaTAM: Splat, Track & Map 3D Gaussians for Dense RGB-D SLAM

Nikhil Keetha, Jay Karhade, Krishna Murthy Jatavallabhula, Gengshan Yang, Sebastian Scherer, Deva Ramanan, Jonathon Luiten
CVPR 2024

[Project](#) / [Paper](#) / [Code](#) / [Video](#)

Explicit 3D Gaussians enable online tracking and dense mapping from a single unposed RGB-D camera, with high-quality scene reconstruction and rendering.

FoundLoc: Vision-based Onboard Aerial Localization in the Wild

Yao He*, Ivan Cisneros*, **Nikhil Keetha**, Jay Patrikar, Zelin Ye, Ian Higgins, Yaoyu Hu, Parv Kapoor, Sebastian Scherer
arXiv 2023

[Project](#) / [Paper](#)

AnyLoc-based place recognition and visual-inertial odometry enable onboard aerial localization against satellite imagery, without GPS or a known starting pose.

AnyLoc: Towards Universal Visual Place Recognition

Nikhil Keetha*, Avneesh Mishra*, Jay Karhade*, Krishna Murthy Jatavallabhula, Sebastian Scherer, Madhava Krishna, Sourav Garg

IEEE RA-L 2024 · ICRA 2024

[Project](#) / [Paper](#) / [Code](#) / [Video](#)

Off-the-shelf image foundation model features and unsupervised aggregation enable state-of-the-art visual place recognition across environments, without task-specific training.

Press

MapAnything

- [New Model Helps Robots See in 3D](#). Carnegie Mellon Robotics Institute · October 2025 · Research feature.
- [The End of VGGT? Why MapAnything Wins in the Real World](#). World of Robot Vision · Technical article.

Meta Hyperscape

- [Share Your Space: Invite Friends to Hang Out in Your Own Hyperscape Worlds](#). Meta Quest Blog · November 2025 · Product feature.
- [Hyperscape Capture](#). Jonathon Luiten · September 2025 · Public demo.
- [Introducing Hyperscape](#). Jonathon Luiten · September 2024 · Public demo.

ViSafe

- [AI-Powered Vision System Helps Drones Navigate Safely](#). Carnegie Mellon University · July 2025 · Research feature.
- [ViSafe: Smarter Vision for Safer Skies](#). Carnegie Mellon Robotics Institute · June 2025 · Research feature.
- [Vision-based aircraft collision avoidance](#). Popular Science · Press coverage.

UFM

- [Unified dense correspondence in action](#). Bilawal Sidhu · June 2025 · Community feature.
- [UFM video feature](#). Instagram · Video.

Map It Anywhere

- [Map It Anywhere with Voxel51](#). Voxel51 · Video feature.

SplaTAM

- [SplaTAM: SLAM with Speed, Precision, and Gaussian Splats](#). Radiance Fields · Technical article.
- [SplaTAM at CMU](#). CMU Extended Reality Technology Center · April 2024 · Research feature.

Talks

Image Matching: Local Features & Beyond

Invited workshop talk · CVPR 2026

Jun 2026

Learning 3D from Multi-view

Guest lecture · CMU 16-825, Learning for 3D Vision [Slides](#)

Nov 2025

Advanced Computer Vision

Guest lecture · CMU 16-820

Fall 2025

MapAnything

Invited research talks · Amazon FAR, Meta, and Field AI [Project](#)

Fall 2025

SplaTAM

Invited webinar · Augmented Reality for Enterprise Alliance (AREA) [Recording](#)

Fall 2024

SplaTAM

Invited research talks · MIT and Voxel AI [Project](#)

Spring 2024

Projects

Research collaborations at Carnegie Mellon University:

- AIST - Frontier AI Research, Japan
- IARPA - WRIVA
- Near Earth Autonomy
- DARPA - Shared Airspace
- DSTA - Multi-Robot Search & Rescue
- Bosch

Service

Reviewing

2021 - Present

Conferences and workshops: CVPR, ICCV, ECCV, NeurIPS, ICRA, IROS, 3DV, and the Image Matching Workshop (IMW).

Journal: IEEE Robotics and Automation Letters (RA-L).

Teaching

Carnegie Mellon University

Teaching Assistant, 16-824: Visual Learning and Recognition - Spring 2025.

Teaching Assistant, 16-820: Advanced Computer Vision - Fall 2024.

Admissions committees

Carnegie Mellon University

Robotics Institute Graduate Admissions Committee; Robotics Institute Summer Scholars (RISS) Admissions Committee.

Mentoring

Mentored researchers at CMU and research interns at Meta. Participated in the CMU Undergraduate AI Mentoring Program and SCS Mentorship Program.

Outreach

RoboLaunch - Co-organizer

Jan 2022 - Jul 2023

Organized a robotics outreach initiative broadening participation and access to robotics. Led speaker coordination, scheduling, publicity, and streaming; the 2022 series reached over 13,000 students and viewers worldwide.

Tartan Planning Series - Co-organizer

Mar 2023 - May 2023

Organized an interactive series of talks, tutorials, and learning on planning for robotics.

Tartan SLAM Series - Co-organizer

Apr 2021 - Dec 2021

Organized the Summer and Fall editions of an interactive series of talks and tutorials on simultaneous localization and mapping.